Name: Mayank Heda
Subject : ASD&D
Roll no: 2401035
Batch:T12

Experiment 08

Aim To build a CI/CD pipeline on AWS that automatically takes code from a GitHub repository, builds it with AWS CodeBuild, and deploys it to an Amazon S3 bucket hosting a static website, using AWS CodePipeline to orchestrate the stages.

Theory

1. CI/CD

- **Continuous Integration (CI)** means developers frequently merge code into a shared repository. Each merge triggers an automated build and test run, so integration errors are caught early.
- **Continuous Delivery/Deployment (CD)** means every change that passes the build is automatically released to a staging or production environment, without manual deployment steps.
- Together they shorten release cycles, reduce human error and give fast feedback.

2. AWS CodePipeline

- A fully managed continuous delivery service that models the release process as a pipeline of **stages** (Source, Build, Deploy).
- Each stage contains **actions**, and the output of one stage is passed to the next as an **artifact**.
- Artifacts between stages are stored in an S3 artifact bucket, which is why the experiment creates a separate bucket (cicd-artifact-...).
- The pipeline starts automatically when a change is pushed to the source repository.

3. AWS CodeBuild

- A managed build service that compiles source code, runs tests and produces deployable artifacts, with no build servers to maintain.
- It runs inside a container, and you pay only for build time.
- Build instructions are written in a **buildspec.yml** file kept in the repository root. The file defines phases (install, pre_build, build, post_build) and which output files become artifacts.
- CodeBuild needs an **IAM service role** to access other AWS resources.

4. Amazon S3 static website hosting

- S3 stores objects (files) in buckets and can serve HTML, CSS and JavaScript directly as a website.

- Enabling static website hosting gives the bucket a website endpoint, and an index document (index.html) is served as the home page.
- For public access you must turn off "Block public access" and attach a **bucket policy** that allows s3:GetObject to everyone (Principal: *). The experiment does exactly this.

5. GitHub as the source provider

- The repository (aws-cicd-experiment) holds index.html, style.css and buildspec.yml.
- CodePipeline connects to GitHub via OAuth, or a GitHub App connection, and detects new commits on the chosen branch.

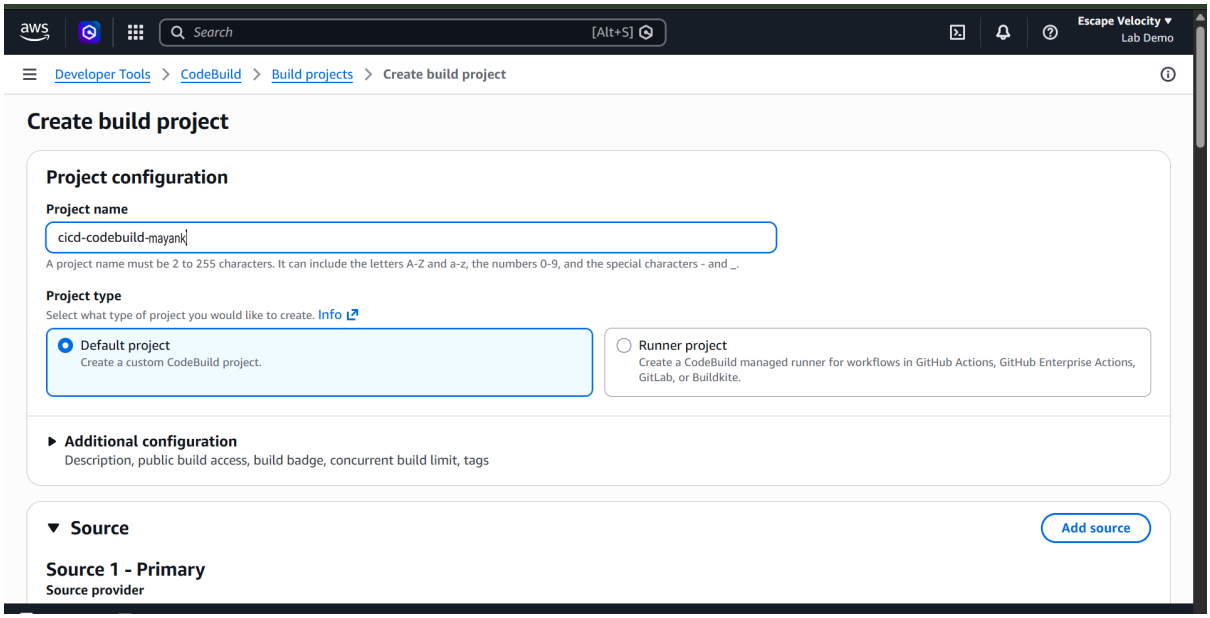
6. IAM roles

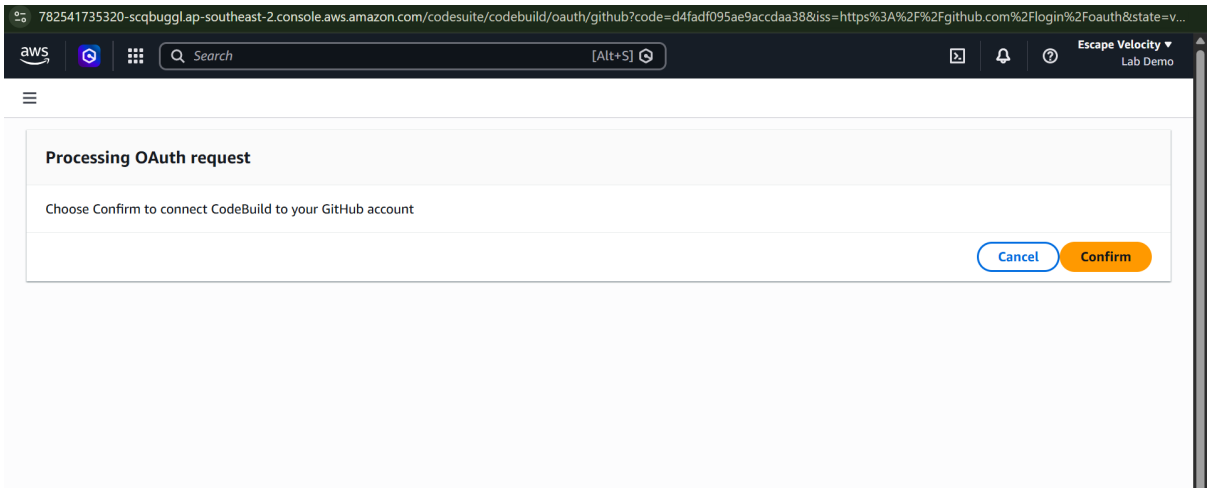
- CodePipeline and CodeBuild each use a service role that grants only the permissions they need, such as reading and writing artifacts in S3 and writing logs.
- This follows the principle of least privilege.

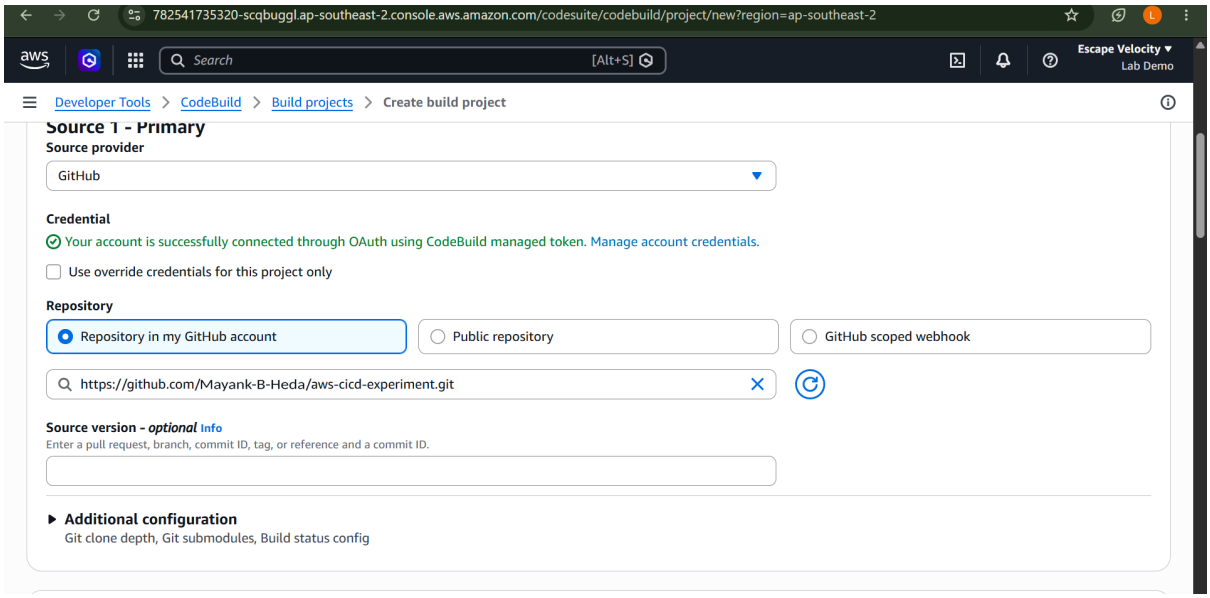
Workflow in this experiment

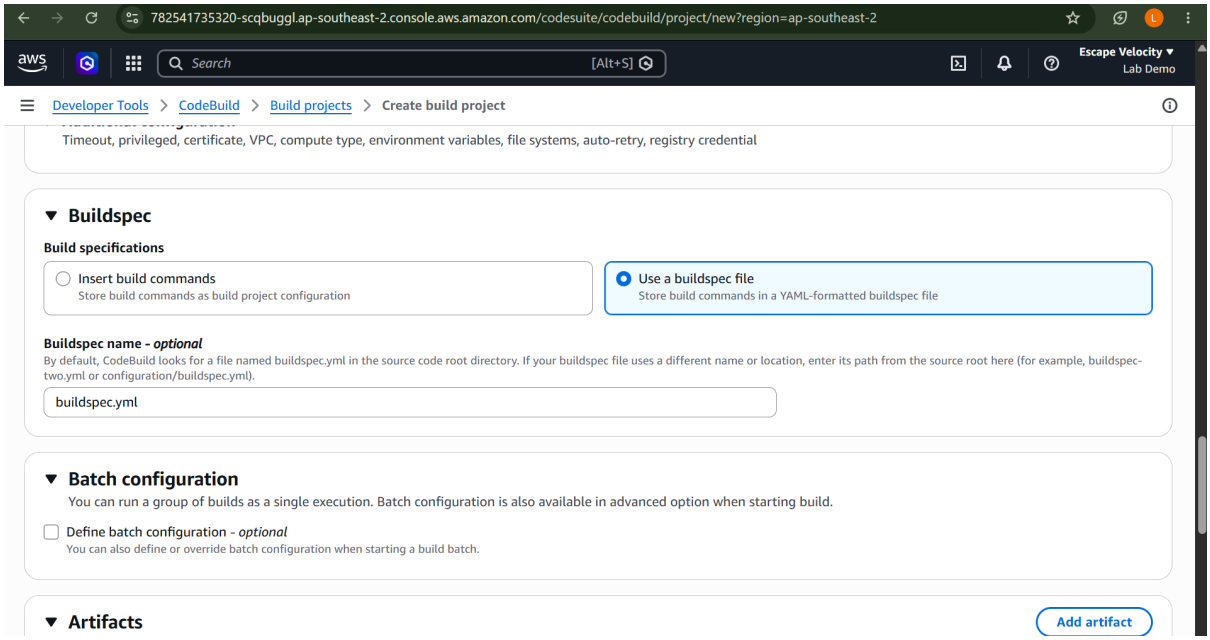
1. Developer pushes a change to GitHub (for example, updating the header in index.html).
2. **Source stage:** CodePipeline pulls the latest code.
3. **Build stage:** CodeBuild runs buildspec.yml and packages the output.
4. **Deploy stage:** The artifact is extracted into the S3 deployment bucket.
5. The updated website is live at the S3 website endpoint, with no manual upload.

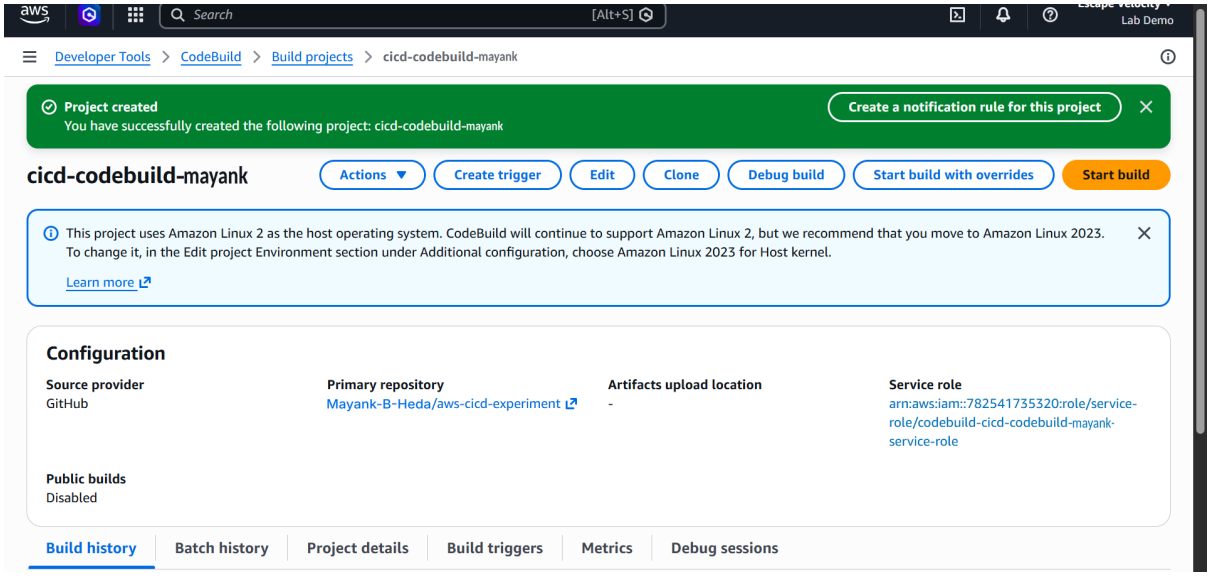
Advantages: automation, faster releases, consistent deployments, fewer manual errors, and easy rollback and traceability.

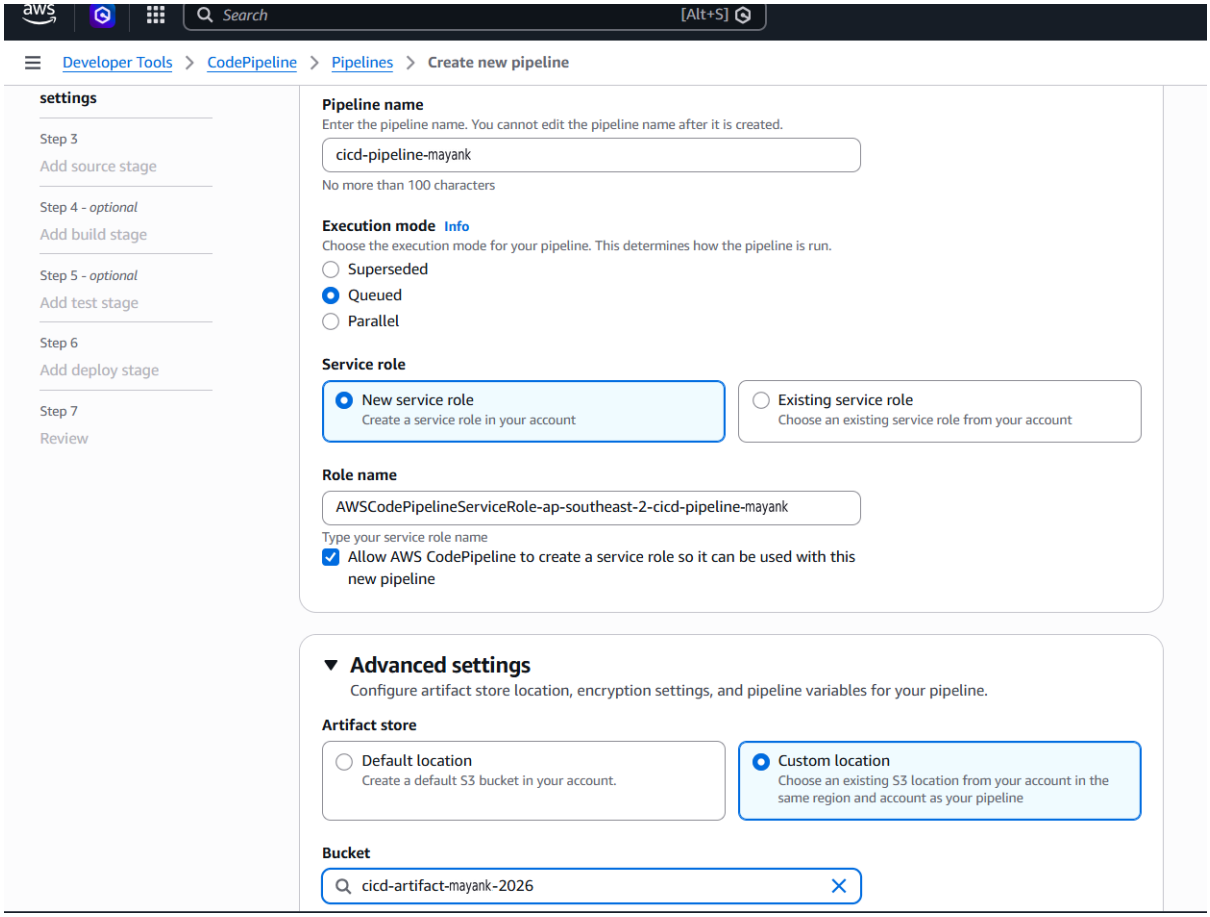

Search
[Alt+S]
Escape Velocity
Lab Demo
Developer Tools
CodeBuild
Build projects
Create build project
Create build project
Project configuration
Project name
cicd-codebuild-mayank
A project name must be 2 to 255 characters. It can include the letters A-Z and a-z, the numbers 0-9, and the special characters - and _.
Project type
Select what type of project you would like to create. Info
Default project
Create a custom CodeBuild project.
Runner project
Create a CodeBuild managed runner for workflows in GitHub Actions, GitHub Enterprise Actions, GitLab, or Buildkite.
Additional configuration
Description, public build access, build badge, concurrent build limit, tags
Source
Add source
Source 1 - Primary
Source provider

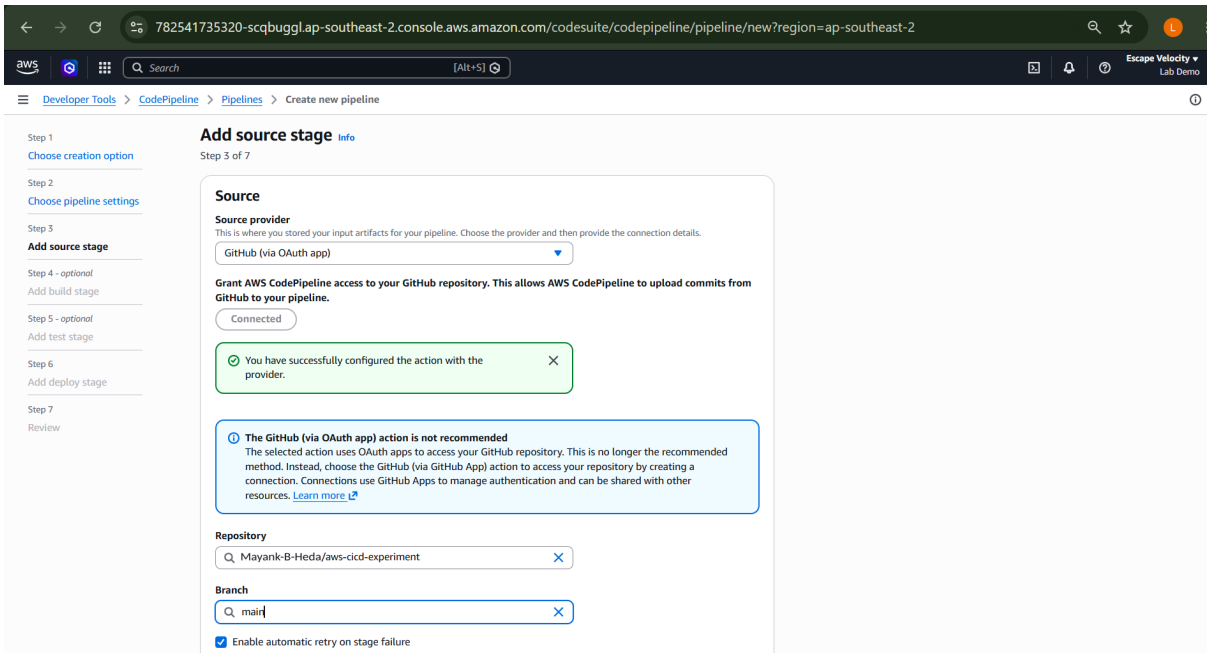

782541735320-scqbuggl.ap-southeast-2.console.aws.amazon.com/codesuite/codebuild/oauth/github?code=d4fadf095ae9accdaa38&iss=https%3A%2F%2Fgithub.com%2Flogin%2Foauth&state=v...
Search
[Alt+S]
Escape Velocity
Lab Demo
Processing OAuth request
Choose Confirm to connect CodeBuild to your GitHub account
Cancel
Confirm

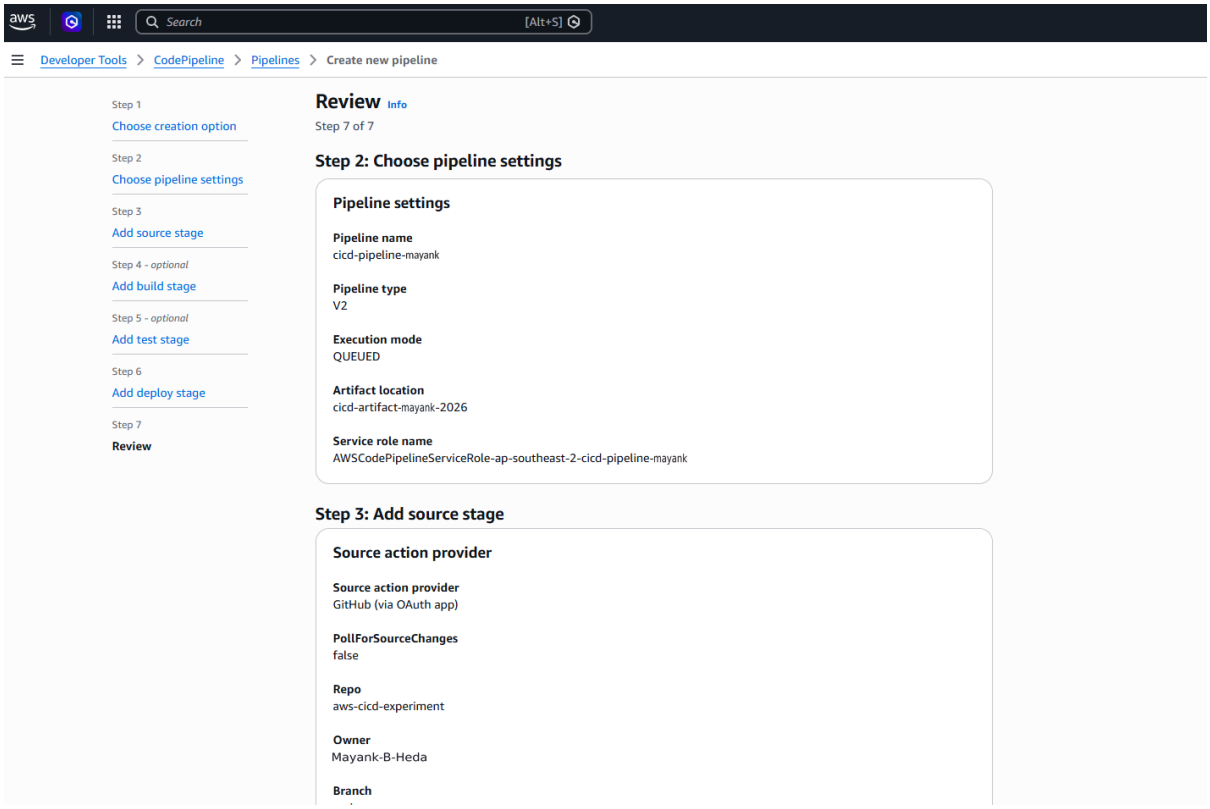

782541735320-scqbuggl.ap-southeast-2.console.aws.amazon.com/codesuite/codebuild/project/new?region=ap-southeast-2
Search
[Alt+S]
Escape Velocity
Lab Demo
Developer Tools
CodeBuild
Build projects
Create build project
Source 1 - Primary
Source provider
GitHub
Credential
Your account is successfully connected through OAuth using CodeBuild managed token. Manage account credentials.
Use override credentials for this project only
Repository
Repository in my GitHub account
Public repository
GitHub scoped webhook
https://github.com/Mayank-B-Heda/aws-cicd-experiment.git
Source version - optional Info
Enter a pull request, branch, commit ID, tag, or reference and a commit ID.
Additional configuration
Git clone depth, Git submodules, Build status config

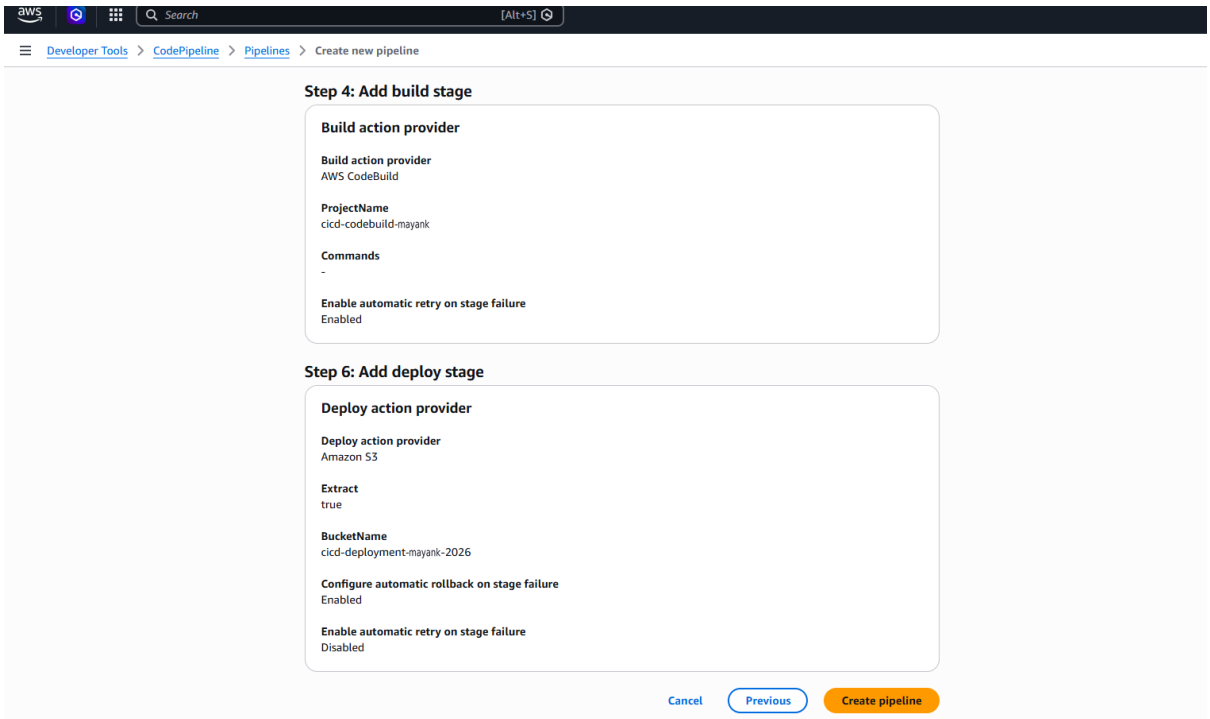

782541735320-scqbuggl.ap-southeast-2.console.aws.amazon.com/codesuite/codebuild/project/new?region=ap-southeast-2
Search
[Alt+S]
Escape Velocity
Lab Demo
Developer Tools > CodeBuild > Build projects > Create build project
Timeout, privileged, certificate, VPC, compute type, environment variables, file systems, auto-retry, registry credential
Buildspec
Build specifications
Insert build commands
Store build commands as build project configuration
Use a buildspec file
Store build commands in a YAML-formatted buildspec file
Buildspec name - optional
By default, CodeBuild looks for a file named buildspec.yml in the source code root directory. If your buildspec file uses a different name or location, enter its path from the source root here (for example, buildspec-two.yml or configuration/buildspec.yml).
buildspec.yml
Batch configuration
You can run a group of builds as a single execution. Batch configuration is also available in advanced option when starting build.
Define batch configuration - optional
You can also define or override batch configuration when starting a build batch.
Artifacts
Add artifact

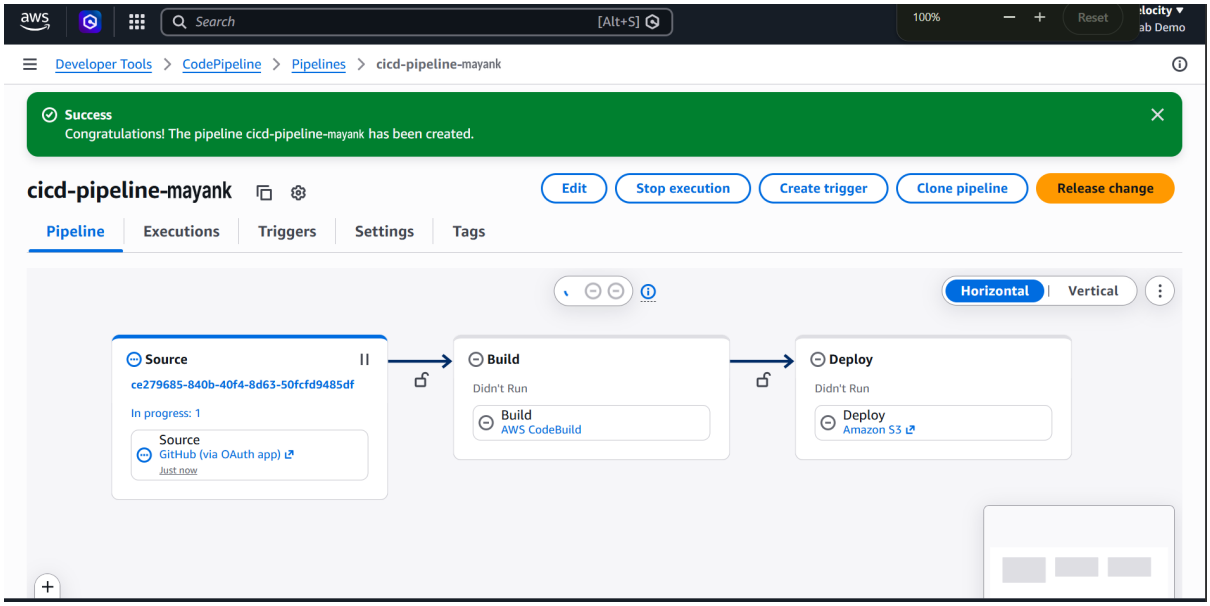

Search
[Alt+S]
Lab Demo
Developer Tools > CodeBuild > Build projects > cicd-codebuild-mayank
Project created
You have successfully created the following project: cicd-codebuild-mayank
Create a notification rule for this project
cicd-codebuild-mayank
Actions
Create trigger
Edit
Clone
Debug build
Start build with overrides
Start build
This project uses Amazon Linux 2 as the host operating system. CodeBuild will continue to support Amazon Linux 2, but we recommend that you move to Amazon Linux 2023.
To change it, in the Edit project Environment section under Additional configuration, choose Amazon Linux 2023 for Host kernel.
Learn more
Configuration
Source provider
GitHub
Primary repository
Mayank-B-Heda/aws-cicd-experiment
Artifacts upload location
-
Service role
arn:aws:iam::782541735320:role/service-role/codebuild-cicd-codebuild-mayank-service-role
Public builds
Disabled
Build history
Batch history
Project details
Build triggers
Metrics
Debug sessions

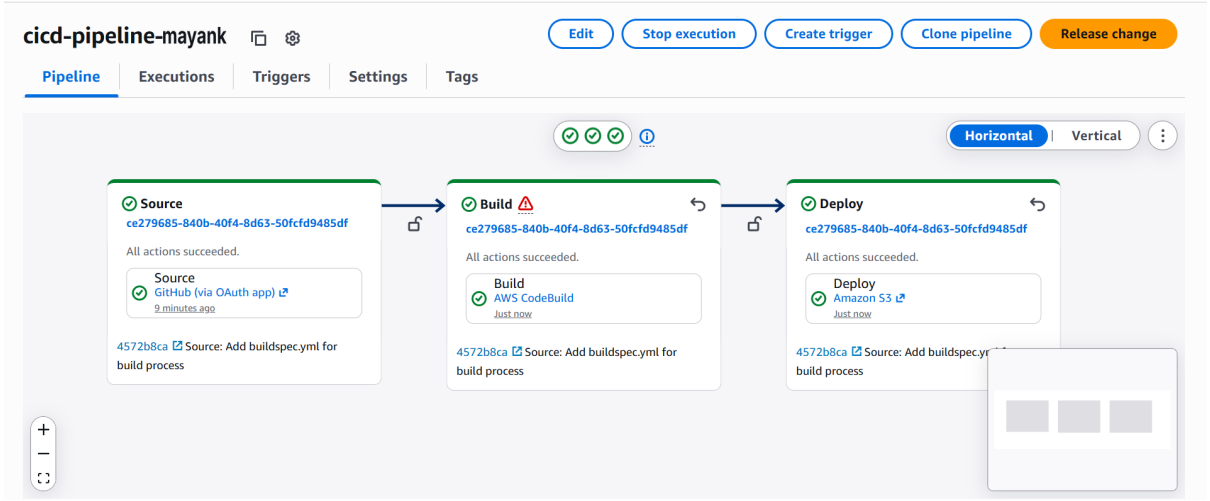

Search
[Alt+S]
Developer Tools > CodePipeline > Pipelines > Create new pipeline
settings
Step 3
Add source stage
Step 4 - optional
Add build stage
Step 5 - optional
Add test stage
Step 6
Add deploy stage
Step 7
Review
Pipeline name
Enter the pipeline name. You cannot edit the pipeline name after it is created.
cicd-pipeline-mayank
No more than 100 characters
Execution mode Info
Choose the execution mode for your pipeline. This determines how the pipeline is run.
Superseded
Queued
Parallel
Service role
New service role
Create a service role in your account
Existing service role
Choose an existing service role from your account
Role name
AWSCodePipelineServiceRole-ap-southeast-2-cicd-pipeline-mayank
Type your service role name
Allow AWS CodePipeline to create a service role so it can be used with this new pipeline
Advanced settings
Configure artifact store location, encryption settings, and pipeline variables for your pipeline.
Artifact store
Default location
Create a default S3 bucket in your account.
Custom location
Choose an existing S3 location from your account in the same region and account as your pipeline
Bucket
cicd-artifact-mayank-2026

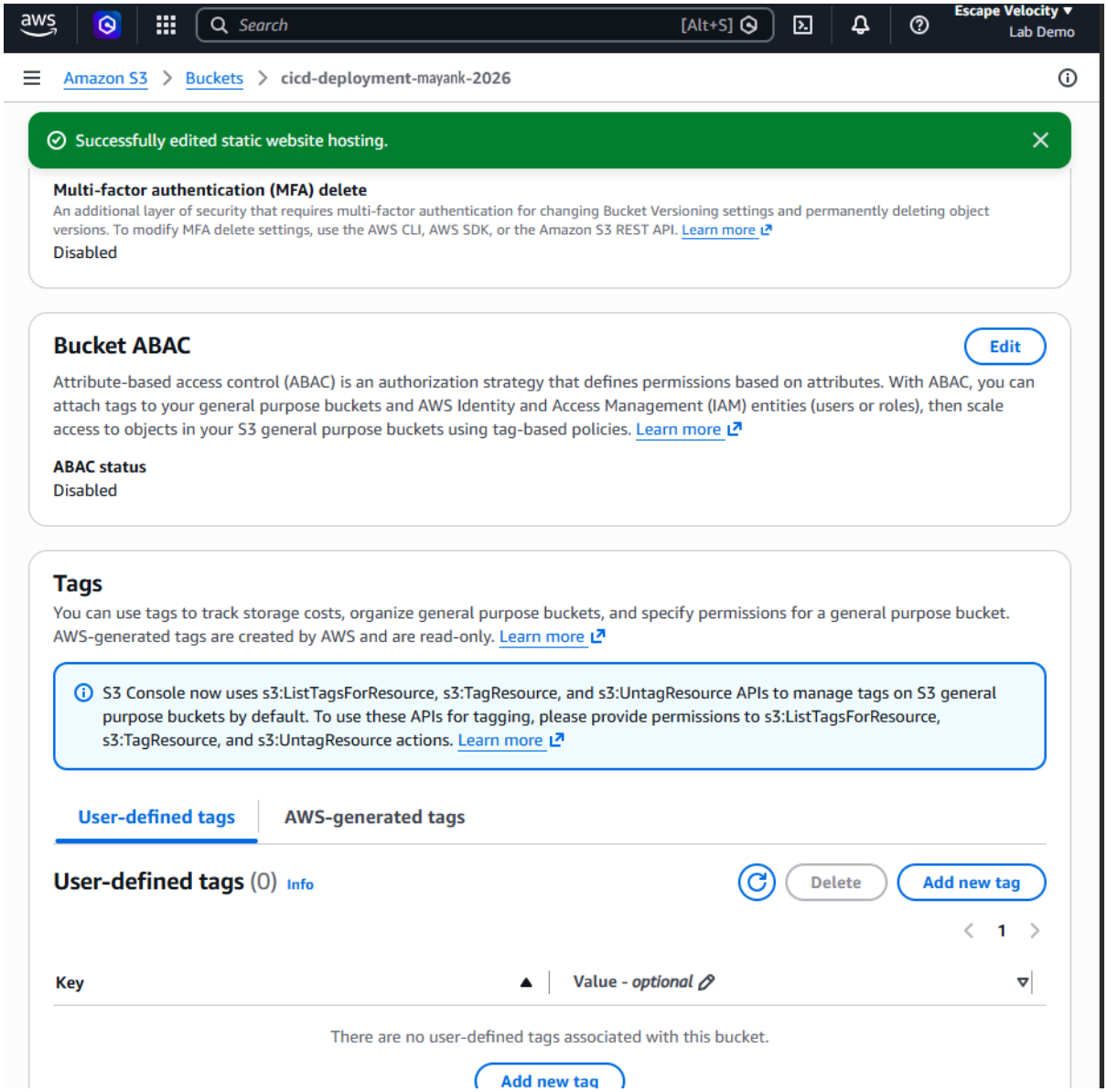

782541735320-scqbuggl.ap-southeast-2.console.aws.amazon.com/codesuite/codepipeline/pipeline/new?region=ap-southeast-2
Search
[Alt+S]
Escape Velocity
Lab Demo
Developer Tools > CodePipeline > Pipelines > Create new pipeline
Step 1
Choose creation option
Step 2
Choose pipeline settings
Step 3
Add source stage
Step 4 - optional
Add build stage
Step 5 - optional
Add test stage
Step 6
Add deploy stage
Step 7
Review
Add source stage Info
Step 3 of 7
Source
Source provider
This is where you stored your input artifacts for your pipeline. Choose the provider and then provide the connection details.
GitHub (via OAuth app)
Grant AWS CodePipeline access to your GitHub repository. This allows AWS CodePipeline to upload commits from GitHub to your pipeline.
Connected
You have successfully configured the action with the provider.
The GitHub (via OAuth app) action is not recommended
The selected action uses OAuth apps to access your GitHub repository. This is no longer the recommended method. Instead, choose the GitHub (via GitHub App) action to access your repository by creating a connection. Connections use GitHub Apps to manage authentication and can be shared with other resources. Learn more
Repository
Mayank-B-Heda/aws-cicd-experiment
Branch
main
Enable automatic retry on stage failure

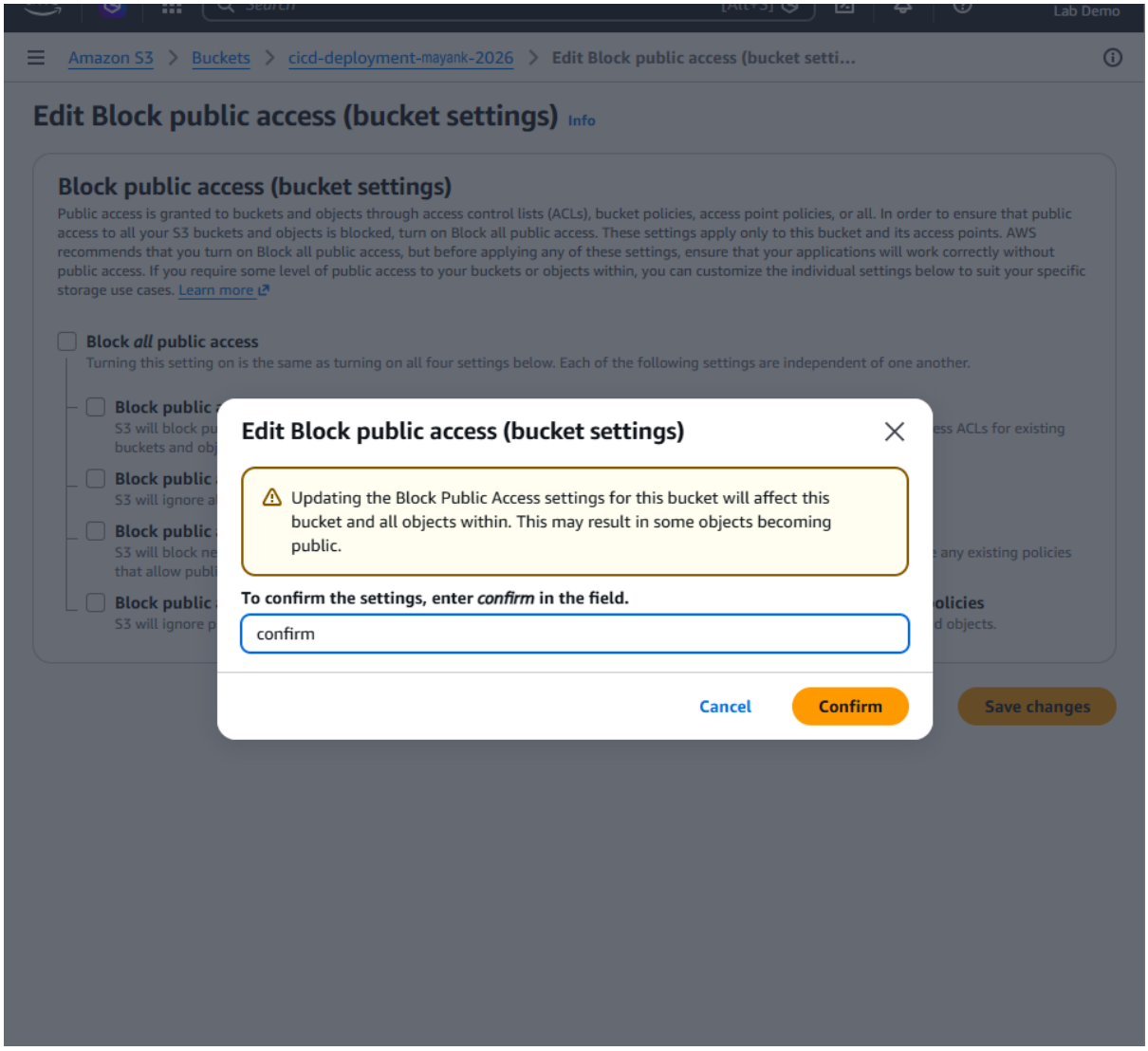

Search
[Alt+S]
Developer Tools
CodePipeline
Pipelines
Create new pipeline
Step 1
Choose creation option
Step 2
Choose pipeline settings
Step 3
Add source stage
Step 4 - optional
Add build stage
Step 5 - optional
Add test stage
Step 6
Add deploy stage
Step 7
Review
Review Info
Step 7 of 7
Step 2: Choose pipeline settings
Pipeline settings
Pipeline name
cicd-pipeline-mayank
Pipeline type
V2
Execution mode
QUEUED
Artifact location
cicd-artifact-mayank-2026
Service role name
AWSCodePipelineServiceRole-ap-southeast-2-cicd-pipeline-mayank
Step 3: Add source stage
Source action provider
Source action provider
GitHub (via OAuth app)
PollForSourceChanges
false
Repo
aws-cicd-experiment
Owner
Mayank-B-Heda
Branch

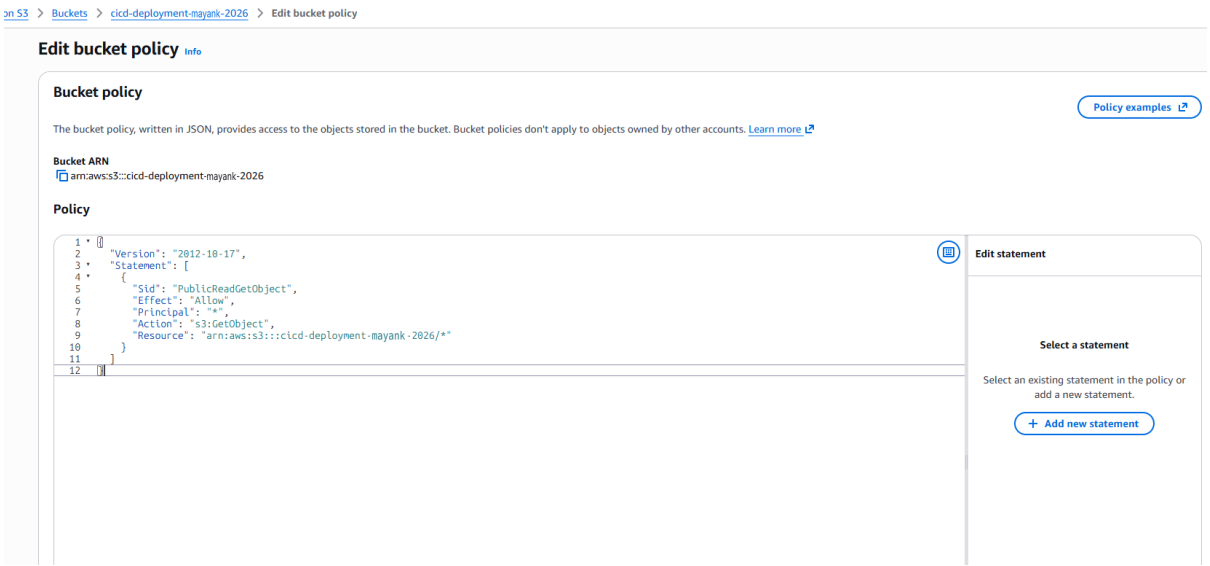

Search
[Alt+S]
Developer Tools
CodePipeline
Pipelines
Create new pipeline
Step 4: Add build stage
Build action provider
Build action provider
AWS CodeBuild
ProjectName
cicd-codebuild-mayank
Commands
-
Enable automatic retry on stage failure
Enabled
Step 6: Add deploy stage
Deploy action provider
Deploy action provider
Amazon S3
Extract
true
BucketName
cicd-deployment-mayank-2026
Configure automatic rollback on stage failure
Enabled
Enable automatic retry on stage failure
Disabled
Cancel
Previous
Create pipeline


Search
[Alt+S]
100%
Reset
Developer Tools
CodePipeline
Pipelines
cicd-pipeline-mayank
Success
Congratulations! The pipeline cicd-pipeline-mayank has been created.
cicd-pipeline-mayank
Edit
Stop execution
Create trigger
Clone pipeline
Release change
Pipeline
Executions
Triggers
Settings
Tags
Horizontal
Vertical
Source
ce279685-840b-40f4-8d63-50fcfd9485df
In progress: 1
Source
GitHub (via OAuth app)
Just now
Build
Didn't Run
Build
AWS CodeBuild
Deploy
Didn't Run
Deploy
Amazon S3


cicd-pipeline-mayank
Edit
Stop execution
Create trigger
Clone pipeline
Release change
Pipeline
Executions
Triggers
Settings
Tags
Horizontal
Vertical
Source
ce279685-840b-40f4-8d63-50fcfd9485df
All actions succeeded.
Source
GitHub (via OAuth app)
9 minutes ago
4572b8ca Source: Add buildspec.yml for build process
Build
ce279685-840b-40f4-8d63-50fcfd9485df
All actions succeeded.
Build
AWS CodeBuild
Just now
4572b8ca Source: Add buildspec.yml for build process
Deploy
ce279685-840b-40f4-8d63-50fcfd9485df
All actions succeeded.
Deploy
Amazon S3
Just now
4572b8ca Source: Add buildspec.yml for build process


Search
[Alt+S]
Escape Velocity
Lab Demo
Amazon S3
Buckets
cicd-deployment-mayank-2026
Successfully edited static website hosting.
Multi-factor authentication (MFA) delete
An additional layer of security that requires multi-factor authentication for changing Bucket Versioning settings and permanently deleting object versions. To modify MFA delete settings, use the AWS CLI, AWS SDK, or the Amazon S3 REST API. Learn more
Disabled
Bucket ABAC
Edit
Attribute-based access control (ABAC) is an authorization strategy that defines permissions based on attributes. With ABAC, you can attach tags to your general purpose buckets and AWS Identity and Access Management (IAM) entities (users or roles), then scale access to objects in your S3 general purpose buckets using tag-based policies. Learn more
ABAC status
Disabled
Tags
You can use tags to track storage costs, organize general purpose buckets, and specify permissions for a general purpose bucket. AWS-generated tags are created by AWS and are read-only. Learn more
S3 Console now uses s3:ListTagsForResource, s3:TagResource, and s3:UntagResource APIs to manage tags on S3 general purpose buckets by default. To use these APIs for tagging, please provide permissions to s3:ListTagsForResource, s3:TagResource, and s3:UntagResource actions. Learn more
User-defined tags
AWS-generated tags
User-defined tags (0) Info
Delete
Add new tag
1
Key
Value - optional
There are no user-defined tags associated with this bucket.
Add new tag


Lab Demo
Amazon S3 > Buckets > cicd-deployment-mayank-2026 > Edit Block public access (bucket setti...
Edit Block public access (bucket settings) Info
Block public access (bucket settings)
Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. Learn more
Block all public access
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.
Edit Block public access (bucket settings)
Updating the Block Public Access settings for this bucket will affect this bucket and all objects within. This may result in some objects becoming public.
To confirm the settings, enter confirm in the field.
confirm
Cancel
Confirm
Save changes

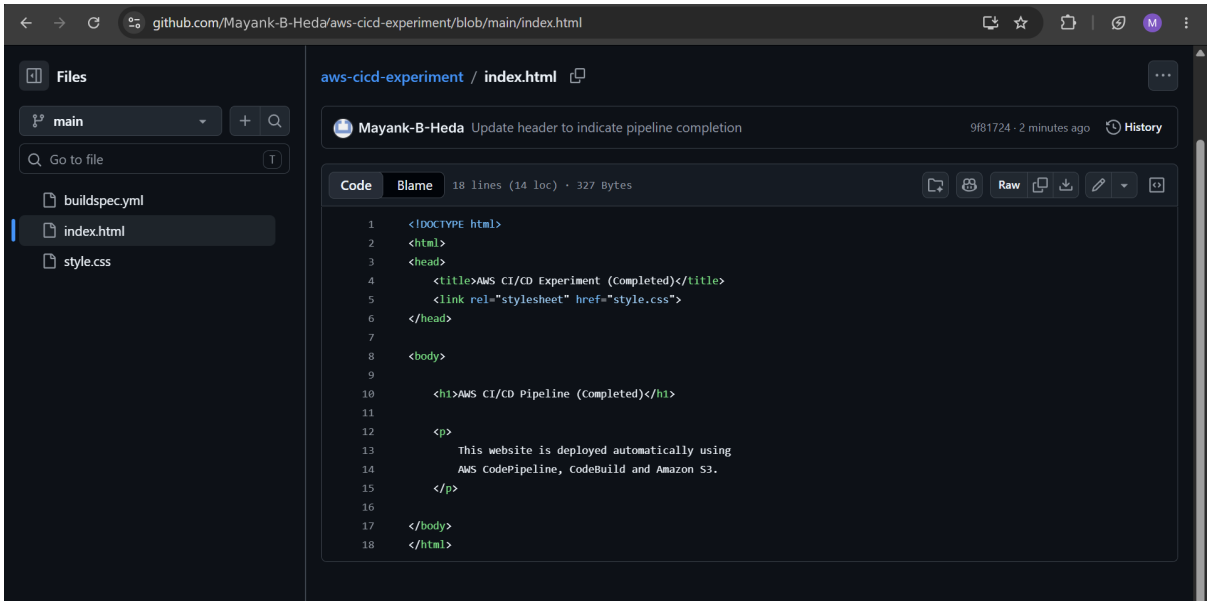

Buckets > cicd-deployment-mayank-2026 > Edit bucket policy
Edit bucket policy Info
Bucket policy
Policy examples
The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. Learn more
Bucket ARN
arn:aws:s3:::cicd-deployment-mayank-2026
Policy
{
"Version": "2012-10-17",
"Statement": [
{
"Sid": "PublicReadGetObject",
"Effect": "Allow",
"Principal": "*",
"Action": "s3:GetObject",
"Resource": "arn:aws:s3:::cicd-deployment-mayank-2026/*"
}
]
}
Edit statement
Select a statement
Select an existing statement in the policy or add a new statement.
+ Add new statement



AWS CI/CD Pipeline

This website is deployed automatically using AWS CodePipeline, CodeBuild and Amazon S3.

After Update:



	Name	Latest execution status	Latest source revisions	Latest execution started	Most recent executions
○	cicd-pipeline-mayank	Succeeded	**Source** – f42e5854: Update title in index.html to indicate completion	1 minute ago	View details



AWS CI/CD Pipeline (Completed)

This website is deployed automatically using AWS CodePipeline, CodeBuild and Amazon S3.

Conclusion

In this experiment we built an automated CI/CD pipeline on AWS using GitHub, CodePipeline, CodeBuild and Amazon S3. A code change pushed to GitHub is built and deployed to the S3 static website without any manual steps. This shows how CI/CD makes software delivery faster, more reliable and less error-prone.